

PHASE-A RESULTS: THE FREE EMA-KL SURROGATE PREVENTS STALENESS COLLAPSE, AND ITS THEORY PICKS THE RIGHT DOSE

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ABSTRACT. Asynchronous RL for LLMs trains on rollouts from stale policy snapshots, and modern recipes increasingly drop KL regularization entirely. A companion theory note shows that a KL penalty anchored on the sampler’s cached log-probabilities, $c(\log T_n - \log T_{n-\Delta})$ with $c = \alpha/(\Delta(1 - \alpha))$, is a zero-cost first-order surrogate for KL regularization against an implicit EMA of the trainer. We test it on Qwen3-4B/Countdown (TBA, no centering) under high and heterogeneous staleness, with and without NVFP4-quantized rollouts. (1) Staleness alone breaks unregularized training — homogeneous $\Delta=32$ falls from eval 0.75 to 0.14 by step 200, and a heterogeneous mixture $\Delta_i \in \{1, 4, 10, 32\}$ collapses ~ 60 steps later ($0.75 \rightarrow 0.002$ by step 300) — while every arm with the surrogate at the theory-prescribed strength is stable, reaching eval 0.79–0.83. (2) Per-rollout calibration $c_i = \alpha/(\Delta_i(1 - \alpha))$ ties the best uniform dose found by a $3\times$ sweep (peaks 0.816 vs. 0.817); that best dose is itself the one the theory transports from a known-good run via the βc invariance, and tripling it costs ~ 1.5 points — the theory selects the right strength without a sweep. (3) The surrogate’s relative error is strongly Δ -dependent as predicted (5.9 at $\Delta \leq 2$ vs. 0.19 at $\Delta \geq 21$), yet training is unaffected. (4) NVFP4 quantization compounds with staleness: the FP4 $\Delta=32$, $\beta=0$ arm collapses ~ 70 steps earlier than its BF16 twin from a lower peak, while $\Delta \lesssim 10$ stays stable; the same theory dose $\beta c = 0.0045$ keeps $\Delta=10$ healthy and extends the $\Delta=32$ collapse boundary $\sim 3\times$, though not to full-horizon stability. In the FP4-heterogeneous cell that motivated this study, the $\beta=0$ control collapses at $\sim$ step 205 while both KL arms train through their full horizons; per-rollout and uniform dosing tie on eval (peaks 0.771 vs. 0.764, single seed) — the rescue claim holds, and the weighting refinement is not separable from a well-chosen constant.

1. INTRODUCTION

Asynchronous RL pipelines for LLMs decouple generation from optimization: inference workers produce rollouts from a policy snapshot that is Δ steps behind the trainer, and the trainer consumes them after the fact. As throughput pressure pushes Δ up — and as quantized inference adds a second source of trainer/sampler mismatch — the question of whether and how to regularize against this drift becomes central. Current practice pulls in two directions: explicit KL regularization against a reference policy costs an extra reference forward pass per batch, so several prominent recipes simply remove the KL term [8], including the humans& NVFP4 reinforcement-learning recipe, which manages mismatch through systems-level bit-exact contracts and trains KL-free [7].

A companion theory note [1] observes that a middle path is already free. In an asynchronous pipeline the sampler’s log-probabilities $\log T_{n-\Delta}$ are cached for importance sampling, so the penalty $c(\log T_n - \log T_{n-\Delta})$ costs nothing extra — and, under a locally-linear-drift assumption, it is a first-order surrogate for KL regularization against an *exponential moving average* of the trainer policy, provided the coefficient is calibrated as $c = \alpha/(\Delta(1 - \alpha))$. The calibration makes two testable predictions: rollouts with different lags Δ_i should receive different weights c_i to maintain a constant effective regularization strength, and a known-good effective dose βc can be transported across staleness regimes without re-tuning.

This report evaluates the surrogate empirically through five questions, each answered by one figure: (i) does staleness alone break unregularized asynchronous RL, and does the free surrogate prevent it (Fig. 1: yes, and yes); (ii) does per-rollout calibration beat a well-chosen constant dose (Fig. 2: a tie when K -groups are age-homogeneous, but the constant itself is the one the

theory prescribes); (iii) is the first-order surrogate accurate everywhere (Fig. 3: no — its error is strongly Δ -dependent, as the theory warns — yet training is robust to it); (iv) does quantization change the picture (Fig. 4: yes — NVFP4 rollouts compound with staleness, moving the $\Delta=32$ collapse boundary from $\sim$ step 170 to $\sim$ 100; the same theory dose fully rescues $\Delta=10$ and delays, but does not eliminate, the $\Delta=32$ failure); (v) does staleness weighting matter under FP4 heterogeneity (Fig. 5: the rescue holds; weighting vs. a constant is a tie at one seed). Except where noted, results are single-seed; tie bands and caveats are stated with each finding.

2. SETUP

All runs use Qwen3-4B-Instruct-2507 on Countdown with TBA (no centering, importance sampling on), $\text{lr } 10^{-6}$, $K=16$, seq. length 4096, 720 target steps, eval every 20 steps, $\alpha = 0.9$, on 2 nodes ($4\times\text{H100}$ inference / $4\times\text{H100}$ training) with 24 h wall-clock allocations (curves end between steps 490 and 690). Full configurations are in Appendix A.

Heterogeneous staleness. `--rl.staleness-offsets` $o_{0..3}$ pins each of 4 data-parallel vLLM ranks to a fixed checkpoint lag (reload every step to step $-o_r$). K -groups stay policy-homogeneous; training batches pool ranks and therefore mix policy ages (verified in the rollout parquets: the trainer’s step-10 batch carries generating checkpoints $\{9, 6, 0, 0\}$). The homogeneous arms use offsets 10, 10, 10, 10 (reference) or the legacy reload rule at `async_level=32` ($\beta=0$ control).

Anchors and calibration. Every arm anchors both the importance ratio and the KL term on the sampler’s own vLLM logprobs. Per-rollout calibration uses each sample’s true lag: $c_i = \alpha/(\Delta_i(1 - \alpha))$, $\Delta_i \geq 1$; the uniform arms apply one dose $w = \beta c_{\text{glob}}$ to every rollout ($c_{\text{glob}} = 0.9/3.2 = 0.28125$). The per-token clamp is in raw log-ratio units (± 10 before the coefficient), so truncation is Δ -invariant.

FP4 emulation. FP4 arms use symmetric NVFP4 QDQ emulation: STE on trainer forwards, QDQ on sampler checkpoints, bit-exact between the two (252 patched linears; embeddings, `lm_head`, and norms stay BF16). The one exception is the asymmetric arm (BF16 trainer, QDQ sampler), reported as a negative result in Section 3.4.

How the uniform constant was chosen (design decision). The uniform arms’ primary constant was fixed by *transporting* the known-good effective strength from the homogeneous $\Delta=10$ baseline ($\beta c = 0.005 \times 0.9 = 0.0045$) and anchoring the global coefficient at the *maximum* staleness: c_{glob} uses $\Delta_{\text{max}} = 32$, and $\beta = 0.0045/0.28125 = 0.016$ so that $\beta c_{\text{glob}} = 0.0045$. This choice does *not* account for the fact that most lags in the $\{1, 4, 10, 32\}$ mixture are below 32: because the raw log-ratio magnitude grows roughly linearly in Δ_i , a constant raw weight yields an effective per-rollout dose $\propto \Delta_i/32$ — sub-maximal lags are effectively under-regularized relative to the calibrated arm, whose effective dose is flat at 0.0045 for every rollout. Alternative constants (the mixture-mean-matched $w=0.015$ and the $\Delta=1$ -matched $w=0.045$) were run in the BF16 heterogeneous sweep (Section 3.2), where the Δ_{max} -anchored $w=0.0045$ performed best; only that point was carried into the FP4 heterogeneous experiments.

3. RESULTS

3.1. Does staleness alone break unregularized async RL? Yes — and the free surrogate prevents it. The homogeneous $\Delta=32$, $\beta=0$ control peaks at eval 0.752 (step 100), then drops sharply between steps 160–180 (train reward $0.68 \rightarrow 0.19$; eval 0.14 at step 200) and oscillates around 0.2–0.45 for 400 more steps without recovering. The heterogeneous $\beta=0$ control — average lag $4\times$ smaller — shows the same failure delayed by ~ 60 steps, preceded by a characteristic train-reward overshoot (0.94 at step 127), then eval 0.002 by step 300. Every arm with the EMA-KL surrogate on, at any tested strength, trains stably through and far beyond these windows (Fig. 1). These runs are BF16 end to end: staleness alone, without quantization, breaks unregularized async RL at these lags, and the zero-cost surrogate fixes it.

As a control on the new machinery itself, the homogeneous $\Delta=10$ reference run through the full pipeline (offsets mechanism, vLLM anchor, raw-units clamp) peaks at 0.829 / final 0.828 — inside the historical no-centering band (0.827–0.831) — so the infrastructure changes do not themselves shift the baseline.



FIGURE 1. Collapse without the KL penalty; stability with it. Yellow (heterogeneous) and violet (homogeneous $\Delta=32$) are $\beta=0$: both collapse (steps ~ 230 and ~ 170) and never recover. Aqua (uniform dose $w=0.0045$) trains through the same window untouched; blue is the homogeneous $\Delta=10$ reference.

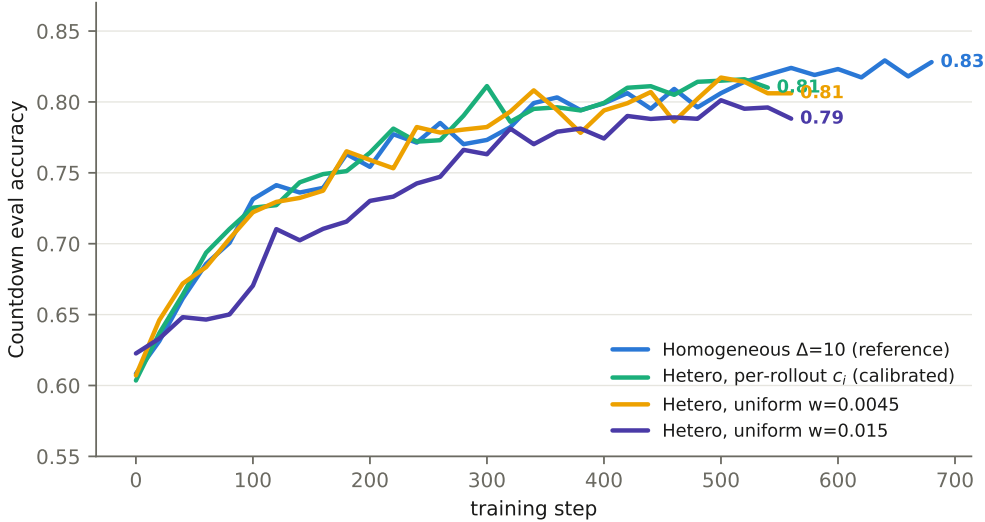


FIGURE 2. All stable arms. Calibrated per-rollout c_i (aqua) vs. the two uniform doses (yellow $w=0.0045$, violet $w=0.015$) under the same $\Delta_i \in \{1, 4, 10, 32\}$ mixture; blue is the homogeneous reference.

3.2. Does per-rollout calibration beat a constant dose? Not here — it ties the best constant, but the best constant is the one the theory picks. Peaks: calibrated 0.816 (step 520), uniform $w=0.0045$ 0.817 (step 500), uniform $w=0.015$ 0.801; homogeneous reference 0.829. The calibrated arm and the best uniform arm are statistically indistinguishable at one seed, and both sit ~ 1 point under the reference. Two facts qualify this tie. First, the best uniform dose is exactly the one the theory prescribes ($w = \beta_{\text{ref}} c_{\text{ref}} = 0.0045$, transported from the known-good homogeneous run via the βc invariance), and tripling it costs 1.5 points: the calibration theory’s practical payoff here is dose selection *without a sweep*, which is what a practitioner changing Δ (or mixing Δ_i) actually needs. Second, the experimental design gives per-rollout weighting little room to act. Because each K -group is policy-homogeneous (all K rollouts in a group share one Δ_i), switching from a constant to a per-rollout weight rescales each group’s summed penalty

by a common per-group factor, and the resulting cross-group differences in dose are diluted by batch-level normalization. Allocation of regularization across rollouts can therefore only matter when policy ages mix *within* a K -group — a configuration this batch does not contain, and the subject of ongoing work (Section 5).

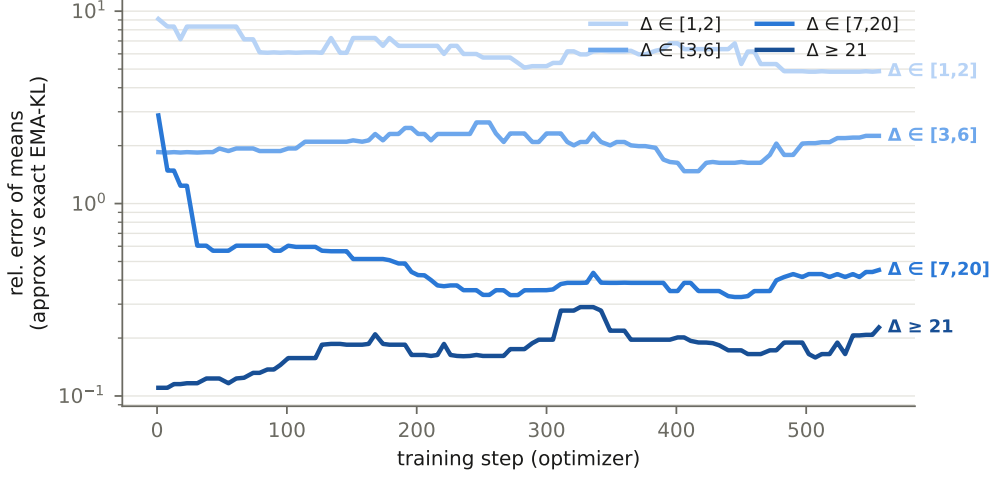


FIGURE 3. Relative error of the surrogate vs. exact EMA-KL by lag stratum (calibrated run; window-median smoothed; log scale).

3.3. Is the first-order surrogate accurate everywhere? No — its error is strongly Δ -dependent, and training does not care. Median relative error of means: 5.88 ($\Delta \leq 2$), 2.13 ($[3, 6]$), 0.48 ($[7, 20]$), 0.19 (≥ 21) — the theory note’s finite-difference variance warning, measured (Fig. 3). Despite a quarter of tokens carrying relative-error- ~ 6 estimates, the calibrated arm matched the reference; the summed, β -scaled penalty is robust to zero-mean estimator noise. This suggests a simple untested refinement: floor Δ_i at ~ 4 in c_i .

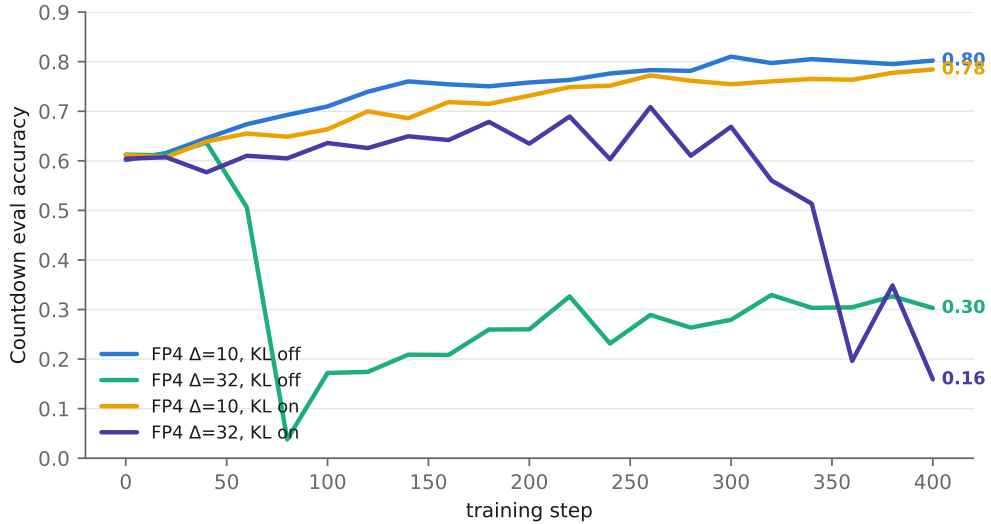


FIGURE 4. FP4 $\beta=0$ ladder and rescue arms. Blue/aqua: $\beta=0$ at $\Delta=10$ (stable) and $\Delta=32$ (collapses by step ~ 100). Yellow/violet: the same theory dose $\beta c = 0.0045$ at $\Delta=10$ (stable through the full horizon) and $\Delta=32$ (holds $\sim 3\times$ past the $\beta=0$ boundary, degrades late).

3.4. Does quantization compound with staleness? Yes — and the same theory dose rescues it, fully at $\Delta=10$ and partially at $\Delta=32$. The $\beta=0$ ladder (Fig. 4): at $\Delta\approx 1$, FP4 training is stable and strong over its 242-step run (train reward 0.80–0.85, eval 0.794 at step 220). At $\Delta=10$ it survives its full horizon (~ 420 steps), volatile in train reward but climbing in eval (0.802 at step 400). At $\Delta=32$ it collapses by step ~ 100 (train reward 0.38; eval already 0.17) and stays collapsed (0.303 at 400). The BF16 twin at the same lag collapsed at step ~ 170 from a higher peak (0.752 vs. 0.637): quantization and staleness *compound* — the boundary moves ~ 70 steps earlier and the pre-collapse peak drops — and at this horizon the $\beta=0$ boundary sits between $\Delta=10$ and 32. The rescue arms apply the *same* theory dose $\beta c = 0.0045$ used throughout, with no quantization-specific retuning. At $\Delta=10$ this is fully sufficient: healthy through the full horizon, eval 0.784 at step 400. At $\Delta=32$ it carries training through and far beyond the twin’s collapse window (0.636 at 100, 0.679 at 180, peak 0.709 at 260 — a $\sim 3\times$ extension of the boundary), but the climb is slower and noisier at reduced altitude, and late in the run evals degrade (0.51 at 340 $\rightarrow$ 0.16 at 400): the unscaled dose delays rather than fully prevents the compounded failure. A quantization-aware dose scaling — treating the quantization gap as extra effective staleness — is the natural refinement, and is untested.

Negative result: the asymmetric arm fails, for an identifiable reason. The asymmetric arm (BF16 trainer, QDQ sampler, KL on at $\beta c = 0.0045$) degrades monotonically from the start: eval 0.56 at step 20, 0.04 at 100, 0.001 by step 180; the run was stopped at step 200. The mechanism is that the surrogate’s persistent quantization gap enters the advantage-level (score-function) penalty un-centered, acting as a global negative reward that suppresses all sampled behavior rather than exerting corrective QAT pressure. A differentiable KL-in-loss path, or centering of the gap, would be needed for this setting.

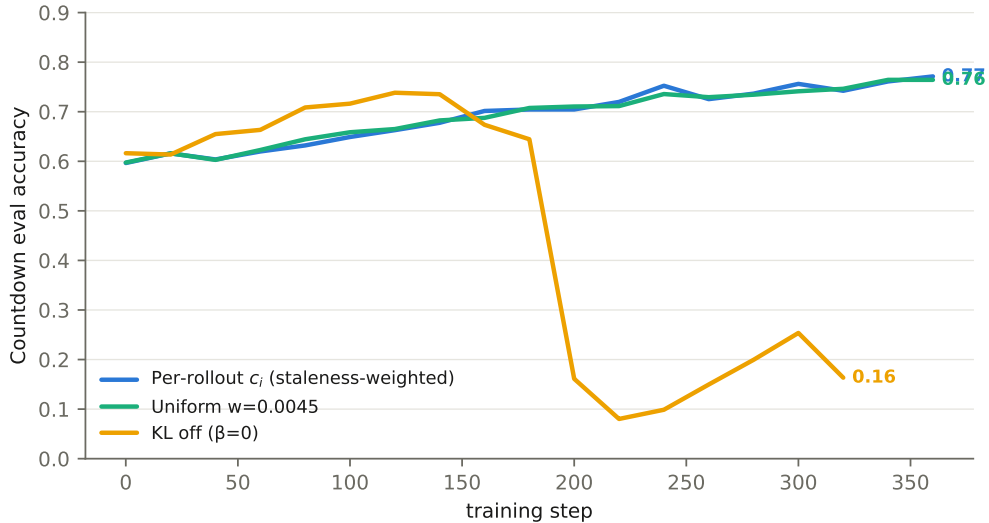


FIGURE 5. The original-hypothesis cell: symmetric NVFP4 with heterogeneous staleness $\Delta_i \in \{1, 4, 10, 32\}$. Yellow ($\beta=0$) collapses at $\sim$ step 205 and never recovers; both KL arms — per-rollout staleness-weighted c_i (blue) and uniform $w=0.0045$ (aqua) — train healthily through their full horizons and end within 0.007 eval of each other.

3.5. Does staleness weighting matter under FP4 heterogeneity? The rescue holds; weighted vs. constant is a tie at one seed. This is the cell that motivated the project: three arms, all symmetric NVFP4 QDQ with the heterogeneous offsets $\Delta_i \in \{1, 4, 10, 32\}$ at `async_level 32` — a $\beta=0$ control, a uniform dose $w=0.0045$ ($\beta=0.016$, global Δ -source), and per-rollout calibrated c_i ($\beta=0.005$) (Fig. 5). The control collapses at $\sim$ step 205 (train reward 0.75 at 140 $\rightarrow$ 0.51 at 200 $\rightarrow$ 0.26 at 212; eval 0.644 at 180 $\rightarrow$ 0.161 at 200) and never recovers through step 327 (eval 0.163 at 320). The boundary at ~ 205 sits between the FP4-homogeneous $\Delta=32$ collapse

(~ 100) and the BF16-heterogeneous collapse (~ 230), consistent with an intermediate effective mismatch. Both KL arms trained healthily through their full horizons (final steps $\sim 361/381$; last evals at step 360); notably, under the heterogeneous mixture the uniform dose did *not* reproduce the late (~ 340) degradation of its homogeneous- $\Delta=32$ counterpart, presumably because the mixture’s mean lag is milder. On eval, the two KL arms tie: peaks 0.771 (per-rollout, step 360) vs. 0.764 (uniform, step 340), finals 0.771 vs. 0.764 at step 360 — both gaps under 0.01 at a single seed, with the lead trading at matched steps (0.705 vs. 0.711 at 200; 0.756 vs. 0.741 at 300; 0.742 vs. 0.746 at 320). Train reward frequently favored the per-rollout arm (e.g. 0.807 vs. 0.735 at 200), but that lead traded on batch noise; we judge on evals. The confirmed part stands: staleness-weighted KL rescues FP4-heterogeneous training where $\beta=0$ collapses; the comparison against uniform dosing is a tie at this seed count, consistent with the age-homogeneous-group mechanism of Section 3.2.

4. RELATED WORK

EMA anchors in RL for LLMs. EMA-PG [2] regularizes policy-gradient training of LLMs against an *explicit* EMA anchor, maintained as a separate model with its own reference forward pass, and treats asynchrony as an orthogonal concern. Our setting obtains an EMA anchor *implicitly*, at zero cost, precisely because of asynchrony: the stale sampler logprobs are already cached, and the theory note [1] shows they support a first-order surrogate for the EMA-KL term. In alignment, TR-DPO [3] and WARP [4] use sequence-level EMA (or averaged) reference policies, updating the anchor toward the trainee during training; these share the moving-anchor intuition but pay for explicit reference evaluations.

KL to the inference policy in async RL. Kimi-K2-style training [5] penalizes divergence from the inference (sampling) policy in an asynchronous pipeline — exactly the object our theory identifies as implicit EMA regularization when the coefficient is held constant. The truncated-importance-sampling line [6] addresses the same trainer/sampler mismatch through the importance-weight path rather than a penalty; the two mechanisms are complementary, and all our arms run with importance sampling on.

KL-free recipes and FP4 training. DAPO [8] removes the KL term entirely, and the humans&NVFP4 recipe [7] trains KL-free (the miles trainer defaults `kl_coef=0`), managing trainer/sampler mismatch through systems-level bit-exact kernel contracts. Our results suggest the free surrogate composes with such recipes: it costs nothing, and it converts high-staleness collapse into stable training. On the numerics side, NVFP4 pretraining [9] and Four-Over-Six [10] develop the quantization formats and scaling rules our QDQ emulation follows.

5. FUTURE WORK

Five directions follow directly from the findings. (1) *Within-group staleness mixing*: the mechanism in Section 3.2 implies per-rollout weighting can only separate from a constant when policy ages mix within a K -group; an experiment with within-group mixing is ongoing. (2) *Quantization-aware dose scaling*: the late degradation of the FP4 $\Delta=32$ rescue suggests treating the quantization gap as extra effective staleness when setting the dose. (3) *The asymmetric setting*: a gap-centered variant or a differentiable KL-in-loss path, so the quantization gap exerts corrective rather than globally negative pressure. (4) *Seeds and tasks*: ≥ 3 paired seeds on calibrated vs. best-uniform, and a second task (MATH). (5) *A Δ_i floor (~ 4) in c_i* , suggested by the stratified error of Section 3.3.

REFERENCES

- [1] B. Bartoldson. Cheaply approximating KL regularization against an EMA. Companion theory note, 2026.
- [2] Zhang and Ba. EMA-PG: policy gradient with an exponential-moving-average anchor. arXiv:2602.04417, 2026.
- [3] Gorbатовski et al. Learn your reference model for real good alignment (TR-DPO). arXiv:2404.09656, 2024.
- [4] Ramé et al. WARP: on the benefits of weight averaged rewarded policies. arXiv:2406.16768, 2024.

- [5] Kimi Team. Kimi K2: open agentic intelligence. arXiv:2507.20534, 2025.
- [6] Yao et al. Your efficient RL framework secretly brings you off-policy RL training. Blog post, <https://fengyao.notion.site/off-policy-rl>, 2025.
- [7] humans&. The 4-bitter lesson. Blog post, July 2026.
- [8] Yu et al. DAPO: an open-source LLM reinforcement learning system at scale. arXiv:2503.14476, 2025.
- [9] NVIDIA. Pretraining large language models with NVFP4. arXiv:2509.25149, 2025.
- [10] Cook et al. Four-Over-Six: NVFP4 quantization scaling. arXiv preprint, 2025.

APPENDIX A. CONFIGURATIONS

Common: model=Qwen/Qwen3-4B-Instruct-2507, LR=1e-6, sampled_k=train_k=16, steps=720, type=tb, conf=COUNTDOWN.toml, seq_length=4096, eval_steps=20, reference_mode=ema, ema_alpha=0.9, kl_centering=false, vllm_logprobs=true, batch 512, wandb fp4_KL, branch fp4_KL.

Arm	β	async	offsets	Δ -source
homog10_approxKL.b005_vllmLP	0.005	10	10,10,10,10	per_rollout
hetero_approxKL.b005_perRollout	0.005	32	1,4,10,32	per_rollout
hetero_approxKL_global.w0045	0.016	32	1,4,10,32	global
hetero_approxKL_global.w015	0.0533	32	1,4,10,32	global
hetero.b0_async32	0	32	1,4,10,32	—
bf16.b0_async32	0	32	— (legacy rule)	—
fp4sym_smoke (60 steps)	0.005	10	—	global
fp4sym.b0_async1	0	1	—	—
fp4sym.b0_async10	0	10	—	—
fp4sym.b0_async32	0	32	—	—
fp4sym_approxKL.b005_async10	0.005	10	—	global
fp4sym_approxKL.b0016_async32	0.016	32	—	global
fp4asym_approxKL.b005_async10 [†]	0.005	10	—	global
fp4hetero.b0_async32	0	32	1,4,10,32	—
fp4hetero_approxKL_global.w0045	0.016	32	1,4,10,32	global
fp4hetero_approxKL.b005_perRollout	0.005	32	1,4,10,32	per_rollout

KL penalty: $-\beta \sum_t c \text{clamp}(\log T_n - \log T_{n-\Delta}, \pm 10)$ added to advantages (no centering); c per-token (c_i) or global per the table. FP4 arms add `rollout_quant=nvfp4 + fake_quant=nvfp4` (252 patched linears; embeddings, `lm_head`, norms BF16). [†]BF16 trainer, QDQ sampler (asymmetric; `fake_quant` off).

APPENDIX B. EVAL TABLE (SELECTED STEPS)

Step	100	200	300	400	500	final
Homog. $\Delta=10$ reference	0.731	0.754	0.773	0.799	0.806	0.828 (680)
Hetero, per-rollout c_i	0.725	0.764	0.811	0.799	0.815	0.810 (540)
Hetero, uniform $w=0.0045$	0.722	0.759	0.782	0.794	0.817	0.806 (560)
Hetero, uniform $w=0.015$	0.670	0.730	0.763	0.774	0.801	0.788 (560)
Hetero, $\beta=0$	0.725	0.662	0.002	0.123	—	0.222 (480)
Homog. $\Delta=32$, $\beta=0$	0.752	0.140	0.246	0.271	0.319	0.340 (540)
FP4 $\Delta \approx 1$, $\beta=0^\ddagger$	0.726	0.786	—	—	—	0.779 (240)
FP4 $\Delta=10$, $\beta=0$	0.710	0.758	0.810	0.802	—	0.802 (400)
FP4 $\Delta=32$, $\beta=0$	0.172	0.260	0.279	0.303	—	0.303 (400)
FP4 $\Delta=10$, KL on	0.664	0.731	0.755	0.784	—	0.784 (400)
FP4 $\Delta=32$, KL on	0.636	0.635	0.669	0.159	—	0.159 (400)
FP4 asym. (BF16 trainer)	0.036	—	—	—	—	0.001 (180)
FP4-hetero, $\beta=0^\ddagger$	0.716	0.161	0.254	—	—	0.163 (320)
FP4-hetero, uniform $w=0.0045$	0.659	0.711	0.741	—	—	0.764 (360)
FP4-hetero, per-rollout c_i	0.649	0.705	0.756	—	—	0.771 (360)

[‡]Run ended early by an infrastructure fault.